Chapter 1: Photosynthesis

Photosynthesis converts light energy into chemical energy in chloroplasts.

Section 1.1 Light reactions: chlorophyll absorbs light, water is split, oxygen is released, ATP and NADPH are produced in the thylakoid membranes.

Section 1.2 Calvin cycle: carbon dioxide is fixed by RuBisCO into sugars in the stroma.

Chapter 2: Cellular respiration

Cells break down glucose to make ATP.

Section 2.1 Glycolysis: glucose is split into two pyruvate in the cytoplasm, net 2 ATP.

Section 2.2 Krebs cycle and electron transport chain: mitochondria produce about 30 ATP, oxygen is the final electron acceptor and water is formed.

Chapter 3: Cell division

Section 3.1 Mitosis: prophase, metaphase, anaphase, telophase produce two identical cells.

Section 3.2 Meiosis: two divisions produce four haploid gametes with genetic variation through crossing over and independent assortment.